



8004 LABS

Your agent's funds, and who can move them

We reviewed the contracts that hold value on 8004 Labs the hard way — then fixed what we found and re-deployed. What changed, what it means for you, and how to check every word against the chain.

What actually holds your agent's money

An 8004 agent is a TaoPunks NFT. That NFT owns a small on-chain account — its own address on Bittensor — and that account is where the agent's TAO balance and staked positions live. Nothing in the platform sits between you and those funds.

You

The holder

Whoever owns the NFT is the sole controller. Only you can send value out of the account.

owns ↓

NFT

The TaoPunk

The token is the key. The account's address is derived from it — no admin key re-points it, and selling the NFT hands over control.

controls ↓

Account

The token-bound account

Holds the balance and the stake. It becomes the agent's Bittensor coldkey directly — funds are never pooled with anyone else's.

may delegate, narrowly ↓

Operator

An optional operator

You can grant an operator a narrow, revocable permission — for example "compound my staking rewards." It can never withdraw. Those bounds are enforced on-chain, not by our app.

That last line is the whole game, and it is exactly what this review set out to prove or break: **can a delegated operator, or anyone other than the holder, get value out of the account?**

HOW WE REVIEWED IT

We tried to break our own custody claim

Passing tests only prove a contract does what its authors expected. So instead of writing more tests, we enumerated the complete set of conditions under which each permission check says *yes* — and looked for sets that grow when they shouldn't, permissions that outlive the person who granted them, and guarantees that turned out to depend on our app rather than the chain.

Four contracts were in scope: the account, the delegation manager, the staking helper, and the task escrow. The review surfaced ten findings. The three that mattered most for custody became a new version of two contracts — shown opposite as before-and-after.

Three fixes, before and after

WHY THE TIMING MATTERED

Bittensor's July runtime upgrade added a new one-transaction function that moves a stake position to another account. A blocklist would have missed it. Because our fix is an allowlist — permit a short safe set, refuse everything else — it blocked the new function automatically. The network's own delegation system moved to the same allowlist model in the same upgrade.

Permissions

An operator's scope can now only be what you last granted

Previously, granting a narrower permission didn't remove the wider one — permissions only ever accumulated.

BEFORE

Re-granting a smaller scope left old permissions live underneath. The only real off-switch was a resettable flag.

AFTER (V2)

Each grant replaces the last exactly. Revoking clears the whole set instantly. What you see is what an operator can do.

Buyers

Buying an agent no longer inherits the seller's hidden control

This is the one that matters most on a marketplace. Delegations, staking policy, and authorizations were all keyed to the agent, not its owner — so they survived a sale.

BEFORE

A buyer silently inherited every operator the seller had ever authorized — potentially the seller themselves, still able to trigger actions.

AFTER (V2)

Every delegation is stamped with the controller who granted it. The moment ownership changes, inherited delegations stop working until the new owner re-grants them.

Withdrawals

Withdrawal functions can never be delegated to an operator — by construction

"Only the holder can move value" used to be a rule our app followed. Now it's enforced in the contract itself.

BEFORE

The contract would technically allow any function to be whitelisted, including stake-movers that redirect funds. Safety rested on the app never doing so.

AFTER (V2)

A fixed allowlist bakes in only the safe, non-withdrawal operations. Every value-exit function — including the new one shipped in the July upgrade — is refused to operators. The holder keeps them all.

THE RESULT

Live, and your stake never moved

The fix was a clean swap. Because an account's address comes from the NFT and never from the manager contract, deploying new manager and staking contracts re-addresses nothing — no funds migrate. The new contracts were deployed, their on-chain bytecode matched against the reviewed source, and the first agent was moved onto them.

A six-point on-chain check confirmed the migration: the new manager is authorized, delegations and staking policy are re-granted under the current owner, and the staked position sits **exactly where it was — still earning**. Retiring the now-inert old manager is the one remaining housekeeping step, reduced to a single click.

- **Your funds are custody-isolated.** Each agent's account is its own Bittensor coldkey; balances are never pooled.
- **Operators are narrow and revocable.** They can compound or manage staking, never withdraw, and you can cut them off in one transaction.
- **A sale is a clean slate.** Buyers no longer inherit a seller's delegations.
- **The rules live on-chain.** These bounds are enforced by verified contracts, not by trusting our servers.

WHAT WE'RE NOT HIDING

The honest limits

BURNING AN AGENT IS PERMANENT AND UNRECOVERABLE

The TaoPunks NFT can be burned. If you burn an agent, its account is bricked and everything inside it — balance and stake — is **frozen forever, with no recovery**. This is confirmed at the chain level; it is not a limitation we can lift for the current collection. The next collection's accounts are being designed to survive it. Until then: never burn an agent that holds funds.

- **Custody is tiered, and TaoPunks is Tier-2.** An automated on-chain check can disqualify a collection but can't fully certify one. TaoPunks carries a burn path and a live admin key, so we describe its custody as trustless derivation with tiered custody — not an unconditional guarantee.
- **A third-party operator you appoint could skim, not steal.** The allowlist stops withdrawals, but not an operator staking your TAO to a validator they control and taking a share of the rewards — principal recoverable, yield divertible. Today every agent is its owner's only operator; a guard ships before third-party operators go live.
- **Independent verification is being opened up.** Everything here is already checkable on-chain. The full method for recounting our completeness proofs will be published — until then, treat "anyone can re-run it" as a commitment, not yet a public fact.

Verify it yourself

The contracts below are live on Bittensor EVM (chain 964) with verified source on the block explorer. You can read every permission and balance directly — no account, no permission, no help from us.

Live, source-verified contracts

MandateManager v2 — delegated execution	0x0B9F5733deda241ecc0141A7B4747fb00c58bb96
AgentStaking v2 — the staking helper	0x0ff30AF8fB38704DC7212f0987CF3343c79916Cf
Agent account implementation	0xfc75cED7B50891043F0CBA8c78882fE46A02EC6A
Agent identity registry	0x5EbB20cbe62a8Ba8A13FDEC06c42b3c1a847ce0e

Read contract state on `lite.chain.opentensor.ai` ; browse verified source and history on `evm.taostats.io` . The only operations grantable to an operator are registration and add/remove-stake — every withdrawal function is refused at the contract level. **ON-CHAIN ENFORCED**

The first migration, at a glance

New delegation manager authorized on the agent's account	Confirmed on-chain
Delegations re-granted, stamped to the current owner	Confirmed on-chain
Staking policy re-set — operator is the owner	Confirmed on-chain
Staked position through the migration	unchanged — 0.9497 a, subnet 64, still earning
Old manager	inert behind the account gate; retirement queued

Three checks anyone can run

THE SOURCE

Read what you'd be trusting

Open `evm.taostats.io` , paste a contract address, and open its Contract tab — the verified source is there to read, line by line.

A PERMISSION

Ask the chain who may act

Call `getMandate` or `isAllowed` with an agent id and an operator address. The chain answers exactly what that operator may do, and until when.

THE MONEY

Check the balance yourself

An agent's account address holds its TAO; staking reads return its live position. Our UI shows the same numbers — you never have to take our word.

8004 Labs — a chain-true agent platform on Bittensor. Custody claims in this report are enforced by verified on-chain contracts and independently checkable at the addresses above. "Build ≠ verify" is our standing rule: nothing here is asserted from a summary — each claim traces to a chain read or a live transaction. Questions or a finding of your own? The contracts are public; bring the transaction. · Report v1.0 · Published July 2026 · 8004.build